

D473

Solutions Design and Visualization Capstone

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A value stream map for a custom cabinet order, drawn by a consultant with a Lean Master certificate, shows total lead time of 480 hours (20 days). The steps customers pay for, cutting, assembly, and finishing, add up to 8 hours. What is the value-added ratio?

- A. About 1.7%
- B. About 98.3%
- C. 60
- D. 40%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A process has a value-added ratio of 5% and 6 hours of value-added time. What is its total lead time?

- A. 0.3 hours
- B. 120 hours
- C. 30 hours
- D. 114 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A fictional e-bike assembler, with a new CFO and a downtown showroom, runs one 8-hour shift with two 15-minute breaks and must build 90 bikes a day. What is its takt time?

- A. About 5.3 minutes
- B. 5.0 minutes
- C. Whatever the slowest station's time is
- D. 0.2 minutes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Takt time is 4 minutes and available time is 440 minutes per shift. How many units per shift does demand require?

- A.** 120
- B.** 1,760
- C.** 4
- D.** 110

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A capstone team is consulting for a fictional snack company whose new product launches take 14 months. The founder started the company in a garage. What should the team produce first in the Define phase?

- A.** A project charter with the problem, a goal such as cutting launch time to 9 months, scope, CTQs, and stakeholders.
- B.** A control plan.
- C.** Baseline data on every launch step.
- D.** A kanban system for the packaging line.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Supplier A quotes \$50 per unit with a 2% defect rate; Supplier B quotes \$52 with a 0.2% defect rate. Each defect costs \$900 to handle. Supplier A's rep offered free trade-show tickets. Annual volume is 10,000 units. Which has the lower total annual cost?

- A.** A: \$500,000 vs. \$520,000.
- B.** A, given the goodwill from the tickets.
- C.** B, saving \$162,000.
- D.** B: \$538,000 vs. \$680,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

In the fictional company's monthly S&OP meeting, sales wants 20% more finished goods for a promotion, but the plant is at 95% of capacity. The CFO joined last month from a bank. What should the executive S&OP step do?

- A. Leave the decision to the weekly scheduling meeting.
- B. Approve the operations number and drop the promotion without analysis.
- C. Compare costed options (overtime, pre-building, a co-packer, a smaller promotion) and commit to one plan.
- D. Approve the sales number as is.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

The fictional company holds 40,000 units of safety stock across 4 regional distribution centers. The team proposes consolidating into 1 central DC. The DCs are leased from a real estate trust. By the square root law, how much safety stock would the central DC need?

- A. 10,000 units
- B. 40,000 units
- C. 20,000 units
- D. 80,000 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A firm has 1 central DC holding 15,000 units of safety stock and replaces it with 9 regional DCs. By the square root law, what does total safety stock become?

- A. 5,000 units
- B. 45,000 units
- C. 15,000 units
- D. 135,000 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A work cell uses 200 parts per hour. Replenishment lead time is 3 hours, each container holds 50 parts, and the safety factor is 10%. The containers are blue plastic bins. How many kanban containers are needed?

- A. 13
- B. 12
- C. 660
- D. 14

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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11. [TRANSFER — reversed stem / different variable]

A line has 10 kanban containers of 40 parts each, a 2-hour lead time, and no safety factor. What hourly usage does this support?

- A. 200 parts per hour
- B. 800 parts per hour
- C. 20 parts per hour
- D. 400 parts per hour

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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12.

The team's capstone report for the fictional company's executives has 40 pages of analysis, a new three-DC network model, and a projected \$2.1 million annual saving, with launch time cut from 14 to 9 months. The team used a teal color theme. How should the executive summary open?

- A. With every analysis, so nothing is missed.
- B. With the company's history.
- C. With the recommendation, the \$2.1 million saving and 5-month faster launch, and the decision needed.
- D. With the methodology and data sources.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A value stream map for a custom cabinet order, ~~drawn by a consultant with a Lean Master certificate,~~ shows total lead time of 480 hours (20 days). The steps customers pay for, cutting, assembly, and finishing, add up to 8 hours. What is the value-added ratio?

Target word: value-added ratio **Decides:** 8 hours / 480 hours.

A	Correct	$8 / 480 \approx 1.7\%$.
B	Adjacent category	The non-value-added share.
C	Inverted operation	$480 / 8$ divides the wrong way.
D	Confusion set	$8 / 20$ mixes hours with days.

2. Answer B Concept 1 · transfer

A process has a value-added ratio of 5% and 6 hours of value-added time. What is its total lead time?

Target word: solve for lead time **Decides:** $6 / 0.05$.

A	Inverted operation	Multiplies 6 by 0.05.
B	Correct	$6 / 0.05 = 120$.
C	Wrong stem number	Multiplies 6 by 5.
D	Adjacent category	The non-value-added time only.

3. Answer B Concept 2

A fictional e-bike assembler, ~~with a new CFO and a downtown showroom,~~ runs one 8-hour shift with two 15-minute breaks and must build 90 bikes a day. What is its takt time?

Target word: takt time **Decides:** $(480 - 30) / 90$.

A	Dropped element	Uses 480 minutes and drops the breaks.
B	Correct	$450 / 90 = 5.0$.
C	Confusion set	That is the bottleneck cycle time, not takt.
D	Inverted operation	$90 / 450$ is bikes per minute.

4. Answer D Concept 2 · transfer

Takt time is 4 minutes and available time is 440 minutes per shift. How many units per shift does demand require?

Target word: solve for demand **Decides:** $440 / 4$.

A	Wrong stem number	Uses a full 480-minute shift.
B	Inverted operation	Multiplies instead of dividing.
C	Confusion set	Takt is a time, not an output.
D	Correct	$440 / 4 = 110$.

5. Answer A Concept 3

A capstone team is consulting for a fictional snack company whose new product launches take 14 months. ~~The founder started the company in a garage.~~ What should the team produce first in the Define phase?

Target word: first deliverable **Decides:** Define comes before any data or solution.

A	Correct	The charter frames every later phase.
B	Inverted operation	The last phase, not the first.
C	Adjacent category	That is Measure, the next phase.
D	Dropped element	Jumps to a solution without defining the problem.

6. Answer D Concept 4

Supplier A quotes \$50 per unit with a 2% defect rate; Supplier B quotes \$52 with a 0.2% defect rate. Each defect costs \$900 to handle. ~~Supplier A's rep offered free trade show tickets.~~ Annual volume is 10,000 units. Which has the lower total annual cost?

Target word: total cost comparison **Decides:** Defect costs: A $200 \times \$900$; B $20 \times \$900$.

A	Dropped element	Drops defect costs.
B	Camouflage	Gifts are not a cost factor, and accepting them raises ethics issues.
C	Wrong stem number	Counts only the defect-cost gap and ignores B's \$20,000 higher price.
D	Correct	A = 500,000 + 180,000; B = 520,000 + 18,000.

7. Answer C Concept 5

In the fictional company's monthly S&OP meeting, sales wants 20% more finished goods for a promotion, but the plant is at 95% of capacity. ~~The CFO joined last month from a bank.~~ What should the executive S&OP step do?

Target word: executive decision **Decides:** Demand exceeds available capacity.

A	Adjacent category	Scheduling executes plans; it does not resolve demand-supply conflicts.
B	Inverted operation	Swings to the other function without weighing options.
C	Correct	Executives resolve the gap with a single agreed plan.
D	Dropped element	Drops the capacity limit.

8. Answer C Concept 6

The fictional company holds 40,000 units of safety stock across 4 regional distribution centers. The team proposes consolidating into 1 central DC. ~~The DCs are leased from a real estate trust.~~ By the square root law, how much safety stock would the central DC need?

Target word: consolidated safety stock **Decides:** $40,000 / \sqrt{4}$.

A	Confusion set	Divides by 4, a linear assumption.
B	Dropped element	Ignores the pooling effect.
C	Correct	$40,000 / 2 = 20,000$.
D	Inverted operation	Multiplies by $\sqrt{4}$ instead of dividing.

9. Answer B Concept 6 · transfer

A firm has 1 central DC holding 15,000 units of safety stock and replaces it with 9 regional DCs. By the square root law, what does total safety stock become?

Target word: decentralized safety stock **Decides:** $15,000 \times \sqrt{9}$.

A	Inverted operation	Divides by 3.
B	Correct	$15,000 \times 3 = 45,000$.
C	Dropped element	Ignores the loss of pooling.
D	Confusion set	Multiplies by 9, a linear assumption.

10. Answer D Concept 7

A work cell uses 200 parts per hour. Replenishment lead time is 3 hours, each container holds 50 parts, and the safety factor is 10%. ~~The containers are blue plastic bins.~~ How many kanban containers are needed?

Target word: number of kanbans **Decides:** $200 \times 3 \times 1.1 / 50 = 13.2$.

A	Adjacent category	Rounds down, leaving a shortfall.
B	Dropped element	Leaves out the safety factor.
C	Confusion set	Counts parts, not containers.
D	Correct	13.2 rounds up to 14.

11. Answer A Concept 7 · transfer

A line has 10 kanban containers of 40 parts each, a 2-hour lead time, and no safety factor. What hourly usage does this support?

Target word: solve for demand **Decides:** $10 \times 40 / 2$.

A	Correct	400 parts cover 2 hours.
B	Inverted operation	Multiplies by lead time.
C	Wrong stem number	Divides the container size by lead time.
D	Dropped element	Ignores the lead time.

12. Answer C Concept 8

The team's capstone report for the fictional company's executives has 40 pages of analysis, a new three-DC network model, and a projected \$2.1 million annual saving, with launch time cut from 14 to 9 months. ~~The team used a teal color theme.~~ How should the executive summary open?

Target word: opening **Decides:** Executives need the recommendation and impact first.

A	Absolute language	A summary must select.
B	Adjacent category	Background is not the message.
C	Correct	Leads with what leaders must know and do.
D	Inverted operation	Puts support before the conclusion.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Value-added ratio	Value-added ratio = value-added time / total lead time, in the same units.	Mixing hours and days, or reporting the non-value-added share.	A value stream map with processing time and total lead time.
2	Takt time	Takt time = available working time / customer demand; breaks are not available time.	Using the full shift or confusing takt with cycle time.	Shift length, breaks, and daily demand.
3	Define phase and the charter	A Six Sigma project starts with a charter: problem, goal, scope, CTQs, and stakeholders.	Jumping to solutions or data collection first.	“What should the team produce first?”
4	Total cost of ownership in procurement	Compare price plus quality, delivery, and handling costs, not price alone.	Choosing the lowest quote.	Different defect rates with a cost per defect.
5	Executive S&OP decisions	When demand exceeds capacity, evaluate costed scenarios and commit to one plan.	Approving one function’s number or pushing the decision down.	A sales request against a near-full plant.
6	Square root law of inventory	Total safety stock scales with the square root of the number of stocking locations.	Assuming a linear change.	Consolidating or splitting distribution centers.
7	Kanban sizing	Containers = demand × lead time × (1 + safety factor) / container size, rounded up.	Leaving out the safety factor or rounding down.	Usage rate, lead time, container size, and safety factor.
8	Communicating the solution	An executive summary opens with the recommendation, its quantified impact, and the decision needed.	Opening with method or background.	A long report for busy executives.

Course scope (public description): Solution Design and Visualization Capstone guides learners to synthesize and apply the skills learned throughout their business analytics, operations, and supply chain education. This course gives learners the opportunity to solve challenges in procurement, sales and operations planning (S&OP) processes, distribution, logistics and transportation faced by a fictional company. In the course capstone project, learners play the role of an operations and supply chain manager or consultant, redesigning the existing supply chain to implement lean processes and using Six Sigma methodology to improve efficiency and allow the company to bring new products or services to market faster. Learners in this course address a real operations and supply chain problem and design solutions, which they communicate in a report and a presentation. At the end of the course, learners will have an authentic experience they can add to their portfolio and show employers. Working through this capstone helps learners to understand how their knowledge interacts with real situations and how roles fit within the industry. The D472: 21st Century Operations and Supply Chain course is a prerequisite.